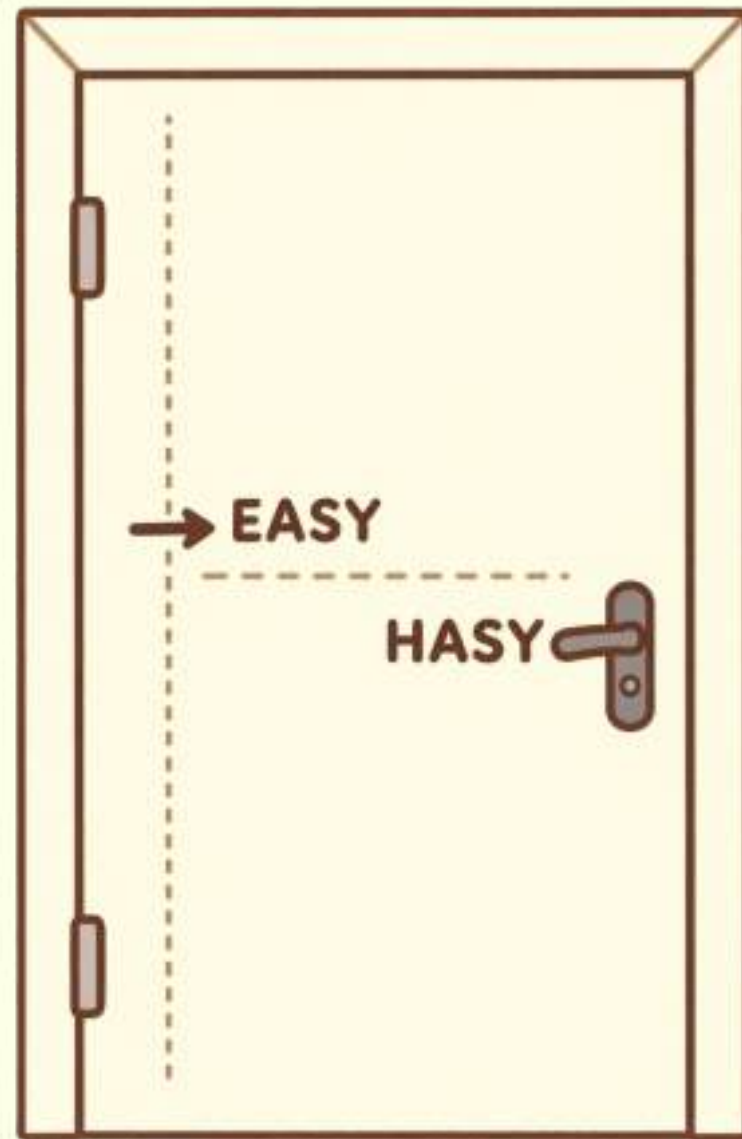




Heavy Lifters — Understanding Torque & Gears



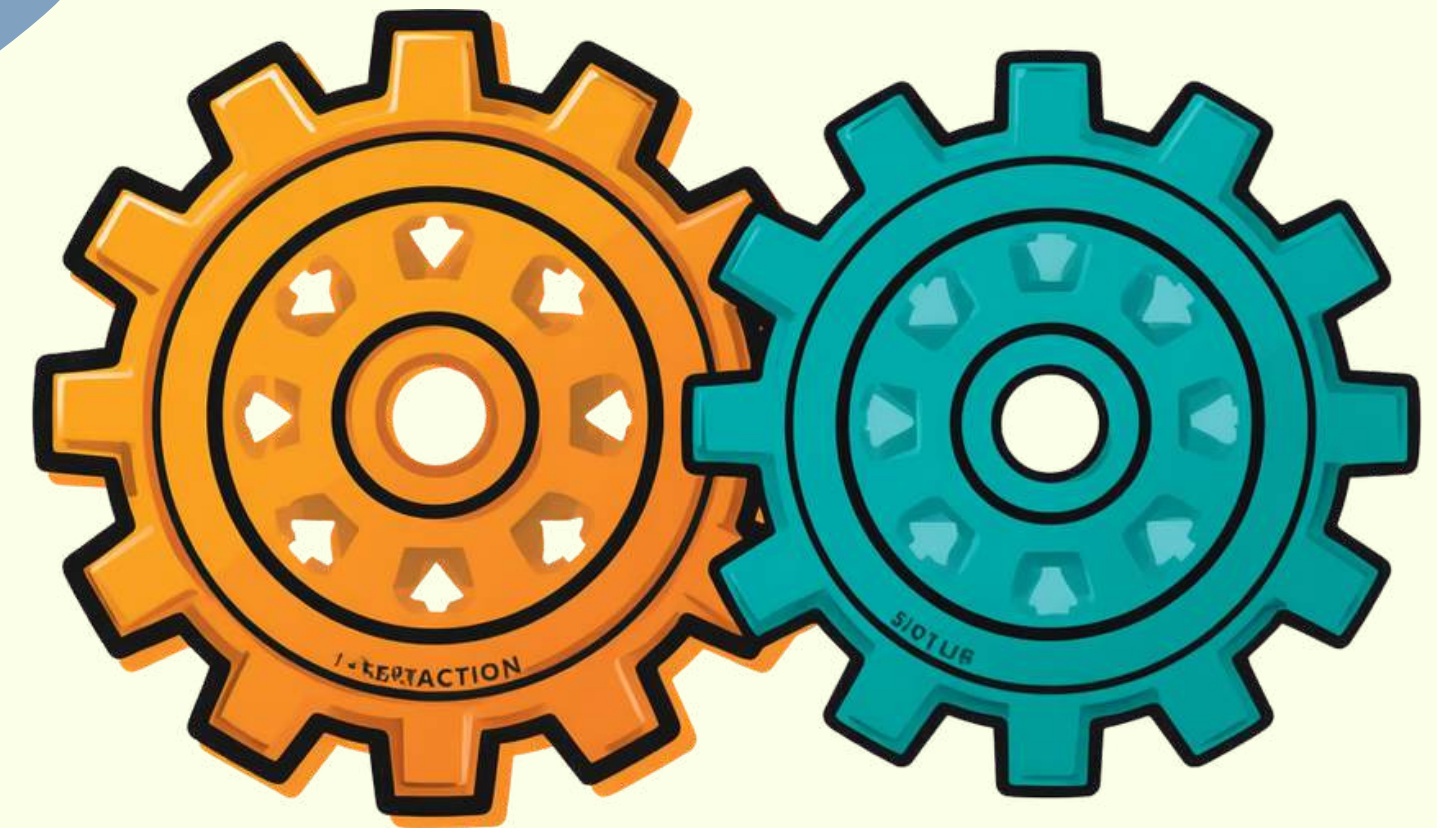
Torque = Turning Force!

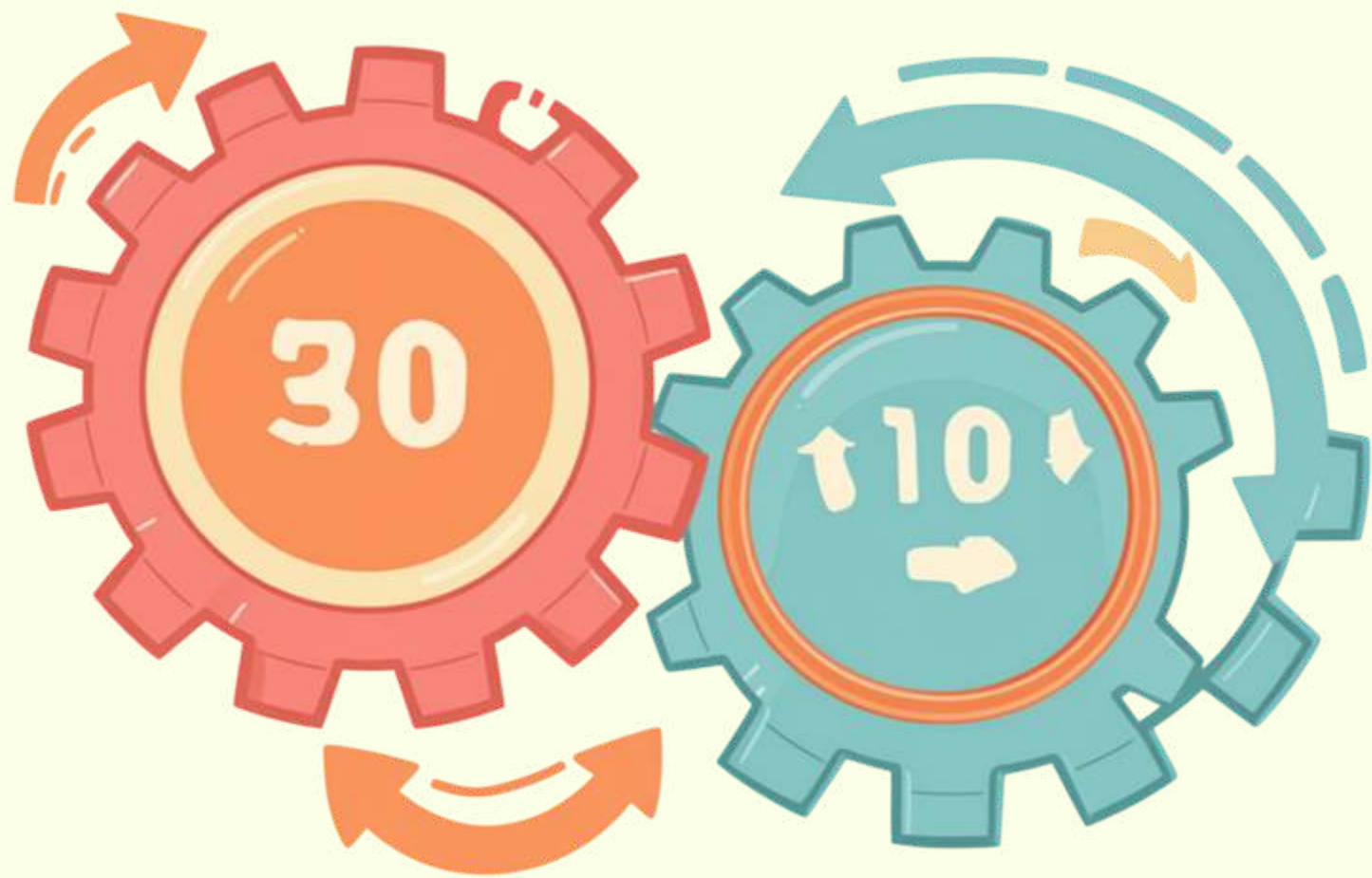
Torque helps us turn heavy things easily!
Push near the edge of a door = easy to open.
Push near the hinge = very hard!
The farther from the center, the more turning force!

Meet the Gears!

Gears help us change speed and power!

- ⚙ Big gear turns slower but stronger
- ⚙ Small gear turns faster but weaker





Gear Ratios Made Simple

🔧 Gear Ratio = Teeth on big gear ÷
Teeth on small gear

📐 Example: $30 \div 10 = 3$

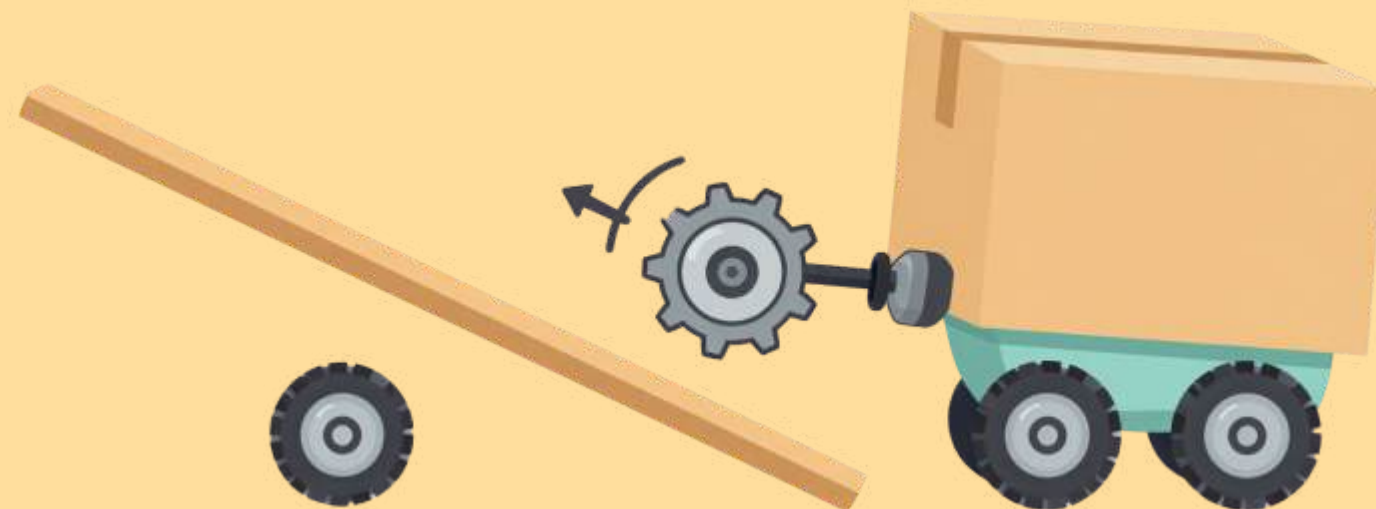
⚙️ 1 turn of big gear = 3 turns of
small gear

Robotics Lab Challenge!

Which gear setup will help lift heavy objects?

Which is faster?

- ⚙️ 1:1 Ratio — Balanced speed & power
- 🚀 1:4 Ratio — Super fast, less power
- 💪 4:1 Ratio — Super strong, lifts heavy loads!



Torque & Gears in Real Life — India Edition



- 🚪 Heavy Gate/Door — long handle = more torque!
- 🌾 Sugarcane Juice Extractor — big gear squeezes hard!
- 🚲 Bicycle on Ghat Road — low gear = more climbing power!
- 🚜 Tractor in Mud — huge torque to pull heavy loads!
- 🌕 Chandrayaan-3 Pragyan Rover — gear ratios to climb lunar terrain!
- 🚖 Auto-rickshaw & Chakki flour mill — gears at work every day!



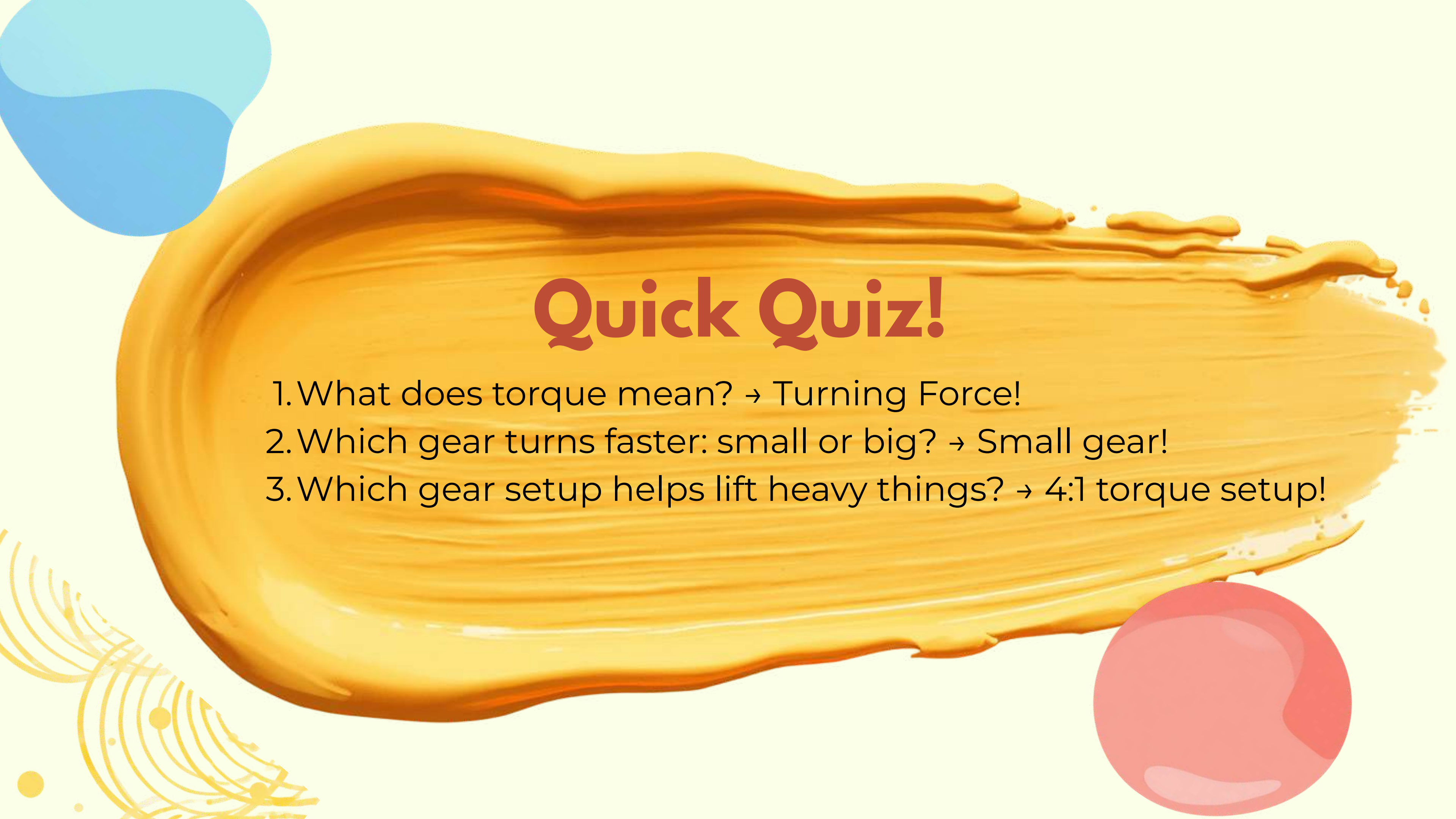
Speed vs Power: What Would You Choose?

⚡ Speed (1:4 ratio): Go fast on a flat road — small gear drives big gear!

💪 Power (4:1 ratio): Pull heavy loads uphill — big gear drives small gear!

Choose speed for racing, choose power for lifting!





Quick Quiz!

1. What does torque mean? → Turning Force!
2. Which gear turns faster: small or big? → Small gear!
3. Which gear setup helps lift heavy things? → 4:1 torque setup!



Thank You!

Keep exploring torque and
gears in your world!
See you in the next session!

Keep turning, keep learning! 

